

Bài thực hành số 1 - Lý thuyết

Xây dựng và triển khai một hợp đồng thông minh cơ bản bằng Vyper

Mục tiêu

- Trình bày được nguyên lý hoạt động của blockchain và smart contract.
 - Hiểu và giải thích các khái niệm cơ bản trong Vyper.
 - Cài đặt và thiết lập môi trường phát triển Vyper.
 - Viết và triển khai một hợp đồng thông minh cơ bản.
 - Kiểm tra và xác minh chức năng của hợp đồng trên testnet.
-

Setup môi trường (Windows)

1. Kiểm tra Python

Nếu máy đã cài Python rồi, kiểm tra phiên bản (cần **3.10+**):

```
In [1]: !python --version
```

Python 3.13.13

Hoặc chạy các cell bên dưới.

```
In [ ]: import platform
import sys
from pathlib import Path

print("Python      :", sys.version)
print("Executable:", sys.executable)
print("Platform   :", platform.platform())
print("CWD         :", Path.cwd())

assert platform.system() == "Windows", "Notebook này chỉ dành cho Windows."
major, minor = sys.version_info[:2]
assert (major, minor) >= (3, 10), f"Can Python >= 3.10, hiện tại {major}.{minor}"
print()
print("[OK] Python / Windows")
```

2. Tạo / kích hoạt virtualenv (PowerShell)

Mở **PowerShell** tại thư mục gốc project:

```
cd d:\Study\IS355
python -m venv .venv
.\.venv\Scripts\Activate.ps1
```

Nếu PowerShell báo lỗi *execution policy*:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

Khi activate thành công, prompt có tiền tố `(.venv)`. Sau đó chọn kernel **Python (IS355 .venv)** (mục 5).

```
In [1]: import sys
from pathlib import Path

prefix = Path(sys.prefix)
in_venv = ('.venv' in prefix.as_posix()) or (sys.prefix != sys.base_prefix)
print('sys.prefix:', sys.prefix)
print('Dang dung venv?', 'CO [OK]' if in_venv else 'KHONG - chay Activate.ps1 roi c

candidates = [
    Path.cwd() / '.venv',
    Path.cwd().parent / '.venv',
    Path.cwd().parent.parent / '.venv',
]
found = next((p for p in candidates if (p / 'Scripts' / 'python.exe').exists()), None)
if found:
    print('Tim thay .venv:', found)
    print('Python trong venv:', found / 'Scripts' / 'python.exe')
else:
    print('Chua co .venv - chay lenh PowerShell o muc 2.')
```

```
sys.prefix : d:\Study\IS355\.venv
Dang dung venv? CO [OK]
Tim thay .venv : d:\Study\IS355\.venv
Python trong venv: d:\Study\IS355\.venv\Scripts\python.exe
```

3. Cài thư viện (download vyper + cryptography)

Package	Mục đích
cryptography	Chữ ký số RSA (mục B)
vyper	Biên dịch hợp đồng <code>.vy</code> trên máy
jupyter / ipykernel	Chạy notebook

PowerShell (đã Activate `.venv`):

```
python -m pip install -U pip
python -m pip install cryptography vyper jupyter ipykernel
```

Có thể dùng thêm [Remix](#) (plugin Vyper) để Deploy trên trình duyệt.

```
In [2]: %pip install -U pip  
%pip install cryptography vyper jupyter ipykernel
```

Requirement already satisfied: pip in d:\Study\IS355\.venv\Lib\site-packages (26.0.1)

Collecting pip

Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)

Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)

```
----- 0.0/1.8 MB ? eta -:--:--
----- 0.0/1.8 MB ? eta -:--:--
----- 0.0/1.8 MB ? eta -:--:--
----- 0.0/1.8 MB ? eta -:--:--
----- 0.3/1.8 MB ? eta -:--:--
----- 0.3/1.8 MB ? eta -:--:--
----- 0.3/1.8 MB ? eta -:--:--
----- 0.5/1.8 MB 480.1 kB/s eta 0:00:03
----- 0.5/1.8 MB 480.1 kB/s eta 0:00:03
----- 0.5/1.8 MB 480.1 kB/s eta 0:00:03
----- 0.8/1.8 MB 418.9 kB/s eta 0:00:03
----- 0.8/1.8 MB 418.9 kB/s eta 0:00:03
----- 0.8/1.8 MB 418.9 kB/s eta 0:00:03
----- 1.0/1.8 MB 415.6 kB/s eta 0:00:02
----- 1.0/1.8 MB 415.6 kB/s eta 0:00:02
----- 1.0/1.8 MB 415.6 kB/s eta 0:00:02
----- 1.0/1.8 MB 415.6 kB/s eta 0:00:02
----- 1.3/1.8 MB 376.6 kB/s eta 0:00:02
----- 1.3/1.8 MB 376.6 kB/s eta 0:00:02
----- 1.3/1.8 MB 376.6 kB/s eta 0:00:02
----- 1.6/1.8 MB 368.7 kB/s eta 0:00:01
----- 1.6/1.8 MB 368.7 kB/s eta 0:00:01
----- 1.6/1.8 MB 368.7 kB/s eta 0:00:01
----- 1.8/1.8 MB 368.4 kB/s 0:00:04
```

Installing collected packages: pip

Attempting uninstall: pip

Found existing installation: pip 26.0.1

Uninstalling pip-26.0.1:

Successfully uninstalled pip-26.0.1

Successfully installed pip-26.2.1

Note: you may need to restart the kernel to use updated packages.

Collecting cryptography

Downloading cryptography-50.0.1-cp311-abi3-win_amd64.whl.metadata (4.3 kB)

Collecting vyper

Using cached vyper-0.4.3-py3-none-any.whl.metadata (7.2 kB)

Collecting jupyter

Using cached jupyter-1.1.1-py2.py3-none-any.whl.metadata (2.0 kB)

Requirement already satisfied: ipykernel in d:\Study\IS355\.venv\Lib\site-packages (7.3.0)

Collecting cffi>=2.0.0 (from cryptography)

Downloading cffi-2.1.1-cp313-cp313-win_amd64.whl.metadata (2.6 kB)

Collecting cbor2<6,>=5.4.6 (from vyper)

Using cached cbor2-5.9.0-cp313-cp313-win_amd64.whl.metadata (5.7 kB)

Requirement already satisfied: asttokens<4,>=2.0.5 in d:\Study\IS355\.venv\Lib\site-packages (from vyper) (3.0.2)

Collecting pycryptodome<4,>=3.5.1 (from vyper)

Using cached pycryptodome-3.23.0-cp37-abi3-win_amd64.whl.metadata (3.5 kB)

Requirement already satisfied: packaging>=23.1 in d:\Study\IS355\.venv\Lib\site-packages (from vyper) (26.3)

Collecting lark<2,>=1.0.0 (from vyper)

Using cached lark-1.3.1-py3-none-any.whl.metadata (1.8 kB)
Collecting wheel (from vyper)
Using cached wheel-0.48.0-py3-none-any.whl.metadata (2.3 kB)
Collecting immutables (from vyper)
Using cached immutables-0.21-cp313-cp313-win_amd64.whl.metadata (4.7 kB)
Collecting notebook (from jupyter)
Downloading notebook-7.6.3-py3-none-any.whl.metadata (10 kB)
Collecting jupyter-console (from jupyter)
Using cached jupyter_console-6.6.3-py3-none-any.whl.metadata (5.8 kB)
Collecting nbconvert (from jupyter)
Using cached nbconvert-7.17.1-py3-none-any.whl.metadata (8.4 kB)
Collecting ipywidgets (from jupyter)
Downloading ipywidgets-8.1.9-py3-none-any.whl.metadata (2.4 kB)
Collecting jupyterlab (from jupyter)
Downloading jupyterlab-4.6.4-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: comm>=0.1.1 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (0.2.3)
Requirement already satisfied: debugpy>=1.6.5 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (1.8.22)
Requirement already satisfied: ipython>=7.23.1 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (9.17.1)
Requirement already satisfied: jupyter-client>=8.9.0 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (8.10.0)
Requirement already satisfied: jupyter-core!=6.0.*,>=5.1 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (5.9.1)
Requirement already satisfied: matplotlib-inline>=0.1 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (0.2.2)
Requirement already satisfied: nest-asyncio>=1.7.0 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (1.7.2)
Requirement already satisfied: psutil>=5.7 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (7.2.2)
Requirement already satisfied: pyzmq>=25 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (27.2.0)
Requirement already satisfied: tornado>=6.4.1 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (6.5.10)
Requirement already satisfied: traitlets>=5.4.0 in d:\Study\IS355\venv\Lib\site-packages (from ipykernel) (5.16.1)
Collecting pycparser (from cffi>=2.0.0->cryptography)
Using cached pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Requirement already satisfied: colorama>=0.4.4 in d:\Study\IS355\venv\Lib\site-packages (from ipython>=7.23.1->ipykernel) (0.4.6)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in d:\Study\IS355\venv\Lib\site-packages (from ipython>=7.23.1->ipykernel) (1.1.1)
Requirement already satisfied: jedi>=0.18.2 in d:\Study\IS355\venv\Lib\site-packages (from ipython>=7.23.1->ipykernel) (0.20.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in d:\Study\IS355\venv\Lib\site-packages (from ipython>=7.23.1->ipykernel) (3.0.53)
Requirement already satisfied: pygments>=2.14.0 in d:\Study\IS355\venv\Lib\site-packages (from ipython>=7.23.1->ipykernel) (2.21.0)
Requirement already satisfied: stack_data>=0.6.0 in d:\Study\IS355\venv\Lib\site-packages (from ipython>=7.23.1->ipykernel) (0.6.3)
Requirement already satisfied: wcwidth>=0.1.4 in d:\Study\IS355\venv\Lib\site-packages (from prompt_toolkit<3.1.0,>=3.0.41->ipython>=7.23.1->ipykernel) (0.8.4)
Requirement already satisfied: parso<0.9.0,>=0.8.6 in d:\Study\IS355\venv\Lib\site-packages (from jedi>=0.18.2->ipython>=7.23.1->ipykernel) (0.8.7)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\Study\IS355\venv\Lib\si

te-packages (from jupyter-client>=8.9.0->ipykernel) (2.9.0.post0)
Requirement already satisfied: typing-extensions>=4.13.0 in d:\Study\IS355\.venv\Lib\site-packages (from jupyter-client>=8.9.0->ipykernel) (4.16.0)
Requirement already satisfied: platformdirs>=2.5 in d:\Study\IS355\.venv\Lib\site-packages (from jupyter-core!=6.0.*,>=5.1->ipykernel) (4.11.12)
Requirement already satisfied: six>=1.5 in d:\Study\IS355\.venv\Lib\site-packages (from python-dateutil>=2.8.2->jupyter-client>=8.9.0->ipykernel) (1.17.0)
Requirement already satisfied: executing>=1.2.0 in d:\Study\IS355\.venv\Lib\site-packages (from stack_data>=0.6.0->ipython>=7.23.1->ipykernel) (2.2.1)
Requirement already satisfied: pure-eval in d:\Study\IS355\.venv\Lib\site-packages (from stack_data>=0.6.0->ipython>=7.23.1->ipykernel) (0.2.4)
Collecting widgetsnbextension~4.0.16 (from ipywidgets->jupyter)
 Downloading widgetsnbextension-4.0.16-py3-none-any.whl.metadata (1.6 kB)
Collecting jupyterlab_widgets~3.0.17 (from ipywidgets->jupyter)
 Downloading jupyterlab_widgets-3.0.17-py3-none-any.whl.metadata (20 kB)
Collecting async_lru>=1.0.0 (from jupyterlab->jupyter)
 Using cached async_lru-2.3.0-py3-none-any.whl.metadata (7.6 kB)
Collecting httpx<1,>=0.25.0 (from jupyterlab->jupyter)
 Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting jinja2>=3.0.3 (from jupyterlab->jupyter)
 Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting jupyter-builder>=1.0.2 (from jupyterlab->jupyter)
 Downloading jupyter_builder-1.2.3-py3-none-any.whl.metadata (7.8 kB)
Collecting jupyter-lsp>=2.0.0 (from jupyterlab->jupyter)
 Using cached jupyter_lsp-2.3.1-py3-none-any.whl.metadata (1.8 kB)
Collecting jupyter-server<3,>=2.19.0 (from jupyterlab->jupyter)
 Downloading jupyter_server-2.21.1-py3-none-any.whl.metadata (8.5 kB)
Collecting jupyterlab-server<3,>=2.28.0 (from jupyterlab->jupyter)
 Downloading jupyterlab_server-2.28.1-py3-none-any.whl.metadata (5.8 kB)
Collecting notebook-shim>=0.2 (from jupyterlab->jupyter)
 Using cached notebook_shim-0.2.4-py3-none-any.whl.metadata (4.0 kB)
Collecting anyio (from httpx<1,>=0.25.0->jupyterlab->jupyter)
 Downloading anyio-4.15.1-py3-none-any.whl.metadata (4.7 kB)
Collecting certifi (from httpx<1,>=0.25.0->jupyterlab->jupyter)
 Downloading certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Collecting httpcore==1.* (from httpx<1,>=0.25.0->jupyterlab->jupyter)
 Using cached httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Collecting idna (from httpx<1,>=0.25.0->jupyterlab->jupyter)
 Downloading idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting h11>=0.16 (from httpcore==1.*->httpx<1,>=0.25.0->jupyterlab->jupyter)
 Using cached h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting argon2-cffi>=21.1 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
 Using cached argon2_cffi-25.1.0-py3-none-any.whl.metadata (4.1 kB)
Collecting jupyter-events>=0.11.0 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
 Using cached jupyter_events-0.12.1-py3-none-any.whl.metadata (5.8 kB)
Collecting jupyter-server-terminals>=0.4.4 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
 Using cached jupyter_server_terminals-0.5.4-py3-none-any.whl.metadata (5.9 kB)
Collecting nbformat>=5.3.0 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
 Downloading nbformat-5.11.1-py3-none-any.whl.metadata (3.7 kB)
Collecting prometheus-client>=0.9 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
 Downloading prometheus_client-0.26.0-py3-none-any.whl.metadata (2.1 kB)
Collecting pywinpty!=3.0.4,>=2.0.1 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)

Downloading pywinpty-3.0.5-cp313-cp313-win_amd64.whl.metadata (5.9 kB)
Collecting send2trash>=1.8.2 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached send2trash-2.1.0-py3-none-any.whl.metadata (4.1 kB)
Collecting terminado>=0.8.3 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached terminado-0.18.1-py3-none-any.whl.metadata (5.8 kB)
Collecting websocket-client>=1.7 (from jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Downloading websocket_client-1.9.2-py3-none-any.whl.metadata (8.3 kB)
Collecting babel>=2.10 (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Using cached babel-2.18.0-py3-none-any.whl.metadata (2.2 kB)
Collecting json5>=0.9.0 (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Downloading json5-0.15.0-py3-none-any.whl.metadata (37 kB)
Collecting jsonschema>=4.18.0 (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Using cached jsonschema-4.26.0-py3-none-any.whl.metadata (7.6 kB)
Collecting requests>=2.31 (from jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting argon2-cffi-bindings (from argon2-cffi>=21.1->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-win_amd64.whl.metadata (7.5 kB)
Collecting MarkupSafe>=2.0 (from jinja2>=3.0.3->jupyterlab->jupyter)
Using cached markupsafe-3.0.3-cp313-cp313-win_amd64.whl.metadata (2.8 kB)
Collecting attrs>=22.2.0 (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Using cached attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting jsonschema-specifications>=2023.03.6 (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl.metadata (2.9 kB)
Collecting referencing>=0.28.4 (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Using cached referencing-0.37.0-py3-none-any.whl.metadata (2.8 kB)
Collecting rpds-py>=0.25.0 (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
Downloading rpds_py-2026.6.3-cp313-cp313-win_amd64.whl.metadata (4.2 kB)
Collecting python-json-logger>=2.0.4 (from jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Downloading python_json_logger-4.2.0-py3-none-any.whl.metadata (2.9 kB)
Collecting pyyaml>=5.3 (from jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached pyyaml-6.0.3-cp313-cp313-win_amd64.whl.metadata (2.4 kB)
Collecting rfc3339-validator (from jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached rfc3339_validator-0.1.4-py2.py3-none-any.whl.metadata (1.5 kB)
Collecting rfc3986-validator>=0.1.1 (from jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached rfc3986_validator-0.1.1-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting fqdn (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached fqdn-1.5.1-py3-none-any.whl.metadata (1.4 kB)
Collecting isoduration (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached isoduration-20.11.0-py3-none-any.whl.metadata (5.7 kB)
Collecting jsonpointer>1.13 (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
Using cached jsonpointer-3.1.1-py3-none-any.whl.metadata (2.4 kB)
Collecting rfc3987-syntax>=1.1.0 (from jsonschema[format-nongpl]>=4.18.0->jupyter-ev

```

ents>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
  Using cached rfc3987_syntax-1.1.0-py3-none-any.whl.metadata (7.7 kB)
Collecting uri-template (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
  Using cached uri_template-1.3.0-py3-none-any.whl.metadata (8.8 kB)
Collecting webcolors>=24.6.0 (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
  Using cached webcolors-25.10.0-py3-none-any.whl.metadata (2.2 kB)
Collecting beautifulsoup4 (from nbconvert->jupyter)
  Downloading beautifulsoup4-4.15.0-py3-none-any.whl.metadata (3.8 kB)
Collecting bleach!=5.0.0 (from bleach[css]!=5.0.0->nbconvert->jupyter)
  Downloading bleach-6.4.0-py3-none-any.whl.metadata (32 kB)
Collecting defusedxml (from nbconvert->jupyter)
  Using cached defusedxml-0.7.1-py2.py3-none-any.whl.metadata (32 kB)
Collecting jupyterlab-pygments (from nbconvert->jupyter)
  Using cached jupyterlab_pygments-0.3.0-py3-none-any.whl.metadata (4.4 kB)
Collecting mistune<4,>=2.0.3 (from nbconvert->jupyter)
  Downloading mistune-3.3.4-py3-none-any.whl.metadata (1.9 kB)
Collecting nbclient>=0.5.0 (from nbconvert->jupyter)
  Downloading nbclient-0.11.0-py3-none-any.whl.metadata (7.3 kB)
Collecting pandocfilters>=1.4.1 (from nbconvert->jupyter)
  Using cached pandocfilters-1.5.1-py2.py3-none-any.whl.metadata (9.0 kB)
Collecting webencodings (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert->jupyter)
  Downloading webencodings-0.6.1-py3-none-any.whl.metadata (3.4 kB)
Collecting tinycss2>=1.1.0 (from bleach[css]!=5.0.0->nbconvert->jupyter)
  Downloading tinycss2-1.5.1-py3-none-any.whl.metadata (3.0 kB)
Collecting fastjsonschema>=2.15 (from nbformat>=5.3.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
  Downloading fastjsonschema-2.22.2-py3-none-any.whl.metadata (2.1 kB)
Collecting charset_normalizer<4,>=2 (from requests>=2.31->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
  Downloading charset_normalizer-3.5.1-cp313-cp313-win_amd64.whl.metadata (46 kB)
Collecting urllib3<3,>=1.26 (from requests>=2.31->jupyterlab-server<3,>=2.28.0->jupyterlab->jupyter)
  Downloading urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Collecting soupsieve>=1.6.1 (from beautifulsoup4->nbconvert->jupyter)
  Downloading soupsieve-2.9.2-py3-none-any.whl.metadata (4.6 kB)
Collecting arrow>=0.15.0 (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
  Using cached arrow-1.4.0-py3-none-any.whl.metadata (7.7 kB)
Collecting tzdata (from arrow>=0.15.0->isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.19.0->jupyterlab->jupyter)
  Downloading tzdata-2026.4-py2.py3-none-any.whl.metadata (1.4 kB)
Downloading cryptography-50.0.1-cp311-abi3-win_amd64.whl (3.8 MB)
----- 0.0/3.8 MB ? eta -:--:--
----- 0.0/3.8 MB ? eta -:--:--
----- 0.8/3.8 MB 3.5 MB/s eta 0:00:01
----- 2.6/3.8 MB 6.4 MB/s eta 0:00:01
----- 3.8/3.8 MB 7.7 MB/s 0:00:00
Using cached vyper-0.4.3-py3-none-any.whl (403 kB)
Using cached cbor2-5.9.0-cp313-cp313-win_amd64.whl (70 kB)
Using cached lark-1.3.1-py3-none-any.whl (113 kB)
Using cached pycryptodome-3.23.0-cp37-abi3-win_amd64.whl (1.8 MB)
Using cached jupyter-1.1.1-py2.py3-none-any.whl (2.7 kB)
Downloading cffi-2.1.1-cp313-cp313-win_amd64.whl (185 kB)
Using cached immutables-0.21-cp313-cp313-win_amd64.whl (35 kB)

```

```

Downloading ipywidgets-8.1.9-py3-none-any.whl (140 kB)
Downloading jupyterlab_widgets-3.0.17-py3-none-any.whl (217 kB)
Downloading widgetsnbextension-4.0.16-py3-none-any.whl (2.2 MB)
----- 0.0/2.2 MB ? eta -:--:--
----- 2.1/2.2 MB 67.8 MB/s eta 0:00:01
----- 2.2/2.2 MB 31.3 MB/s 0:00:00
Using cached jupyter_console-6.6.3-py3-none-any.whl (24 kB)
Downloading jupyterlab-4.6.4-py3-none-any.whl (17.2 MB)
----- 0.0/17.2 MB ? eta -:--:--
----- 4.5/17.2 MB 22.2 MB/s eta 0:00:01
----- 9.4/17.2 MB 22.5 MB/s eta 0:00:01
----- 13.6/17.2 MB 22.7 MB/s eta 0:00:01
----- 17.2/17.2 MB 20.5 MB/s 0:00:00
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Downloading jupyter_server-2.21.1-py3-none-any.whl (394 kB)
Downloading jupyterlab_server-2.28.1-py3-none-any.whl (60 kB)
Downloading anyio-4.15.1-py3-none-any.whl (132 kB)
Using cached argon2_cffi-25.1.0-py3-none-any.whl (14 kB)
Using cached async_lru-2.3.0-py3-none-any.whl (8.4 kB)
Using cached babel-2.18.0-py3-none-any.whl (10.2 MB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Downloading idna-3.20-py3-none-any.whl (69 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading json5-0.15.0-py3-none-any.whl (36 kB)
Using cached jsonschema-4.26.0-py3-none-any.whl (90 kB)
Using cached attrs-26.1.0-py3-none-any.whl (67 kB)
Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl (18 kB)
Downloading jupyter_builder-1.2.3-py3-none-any.whl (947 kB)
----- 0.0/947.5 kB ? eta -:--:--
----- 947.5/947.5 kB 23.0 MB/s 0:00:00
Using cached jupyter_events-0.12.1-py3-none-any.whl (19 kB)
Using cached jsonpointer-3.1.1-py3-none-any.whl (7.7 kB)
Using cached jupyter_lsp-2.3.1-py3-none-any.whl (77 kB)
Using cached jupyter_server_terminals-0.5.4-py3-none-any.whl (13 kB)
Using cached markupsafe-3.0.3-cp313-cp313-win_amd64.whl (15 kB)
Using cached nbconvert-7.17.1-py3-none-any.whl (261 kB)
Downloading mistune-3.3.4-py3-none-any.whl (66 kB)
Downloading bleach-6.4.0-py3-none-any.whl (165 kB)
Downloading nbclient-0.11.0-py3-none-any.whl (25 kB)
Downloading nbformat-5.11.1-py3-none-any.whl (79 kB)
Downloading fastjsonschema-2.22.2-py3-none-any.whl (27 kB)
Using cached notebook_shim-0.2.4-py3-none-any.whl (13 kB)
Using cached pandocfilters-1.5.1-py2.py3-none-any.whl (8.7 kB)
Downloading prometheus_client-0.26.0-py3-none-any.whl (64 kB)
Downloading python_json_logger-4.2.0-py3-none-any.whl (14 kB)
Downloading pywinpty-3.0.5-cp313-cp313-win_amd64.whl (2.1 MB)
----- 0.0/2.1 MB ? eta -:--:--
----- 2.1/2.1 MB 27.9 MB/s 0:00:00
Using cached pyyaml-6.0.3-cp313-cp313-win_amd64.whl (154 kB)
Using cached referencing-0.37.0-py3-none-any.whl (26 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Downloading charset_normalizer-3.5.1-cp313-cp313-win_amd64.whl (199 kB)
Downloading urllib3-2.8.0-py3-none-any.whl (135 kB)
Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)
Using cached rfc3986_validator-0.1.1-py2.py3-none-any.whl (4.2 kB)

```

```

Using cached rfc3987_syntax-1.1.0-py3-none-any.whl (8.0 kB)
Downloading rpds_py-2026.6.3-cp313-cp313-win_amd64.whl (219 kB)
Using cached send2trash-2.1.0-py3-none-any.whl (17 kB)
Using cached terminado-0.18.1-py3-none-any.whl (14 kB)
Downloading tinycss2-1.5.1-py3-none-any.whl (28 kB)
Using cached webcolors-25.10.0-py3-none-any.whl (14 kB)
Downloading webencodings-0.6.1-py3-none-any.whl (8.7 kB)
Downloading websocket_client-1.9.2-py3-none-any.whl (95 kB)
Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-win_amd64.whl (25 kB)
Downloading beautifulsoup4-4.15.0-py3-none-any.whl (109 kB)
Downloading soupsieve-2.9.2-py3-none-any.whl (37 kB)
Using cached defusedxml-0.7.1-py2.py3-none-any.whl (25 kB)
Using cached fqdn-1.5.1-py3-none-any.whl (9.1 kB)
Using cached isoduration-20.11.0-py3-none-any.whl (11 kB)
Using cached arrow-1.4.0-py3-none-any.whl (68 kB)
Using cached jupyterlab_pygments-0.3.0-py3-none-any.whl (15 kB)
Downloading notebook-7.6.3-py3-none-any.whl (5.5 MB)
----- 0.0/5.5 MB ? eta -:--:--
----- 5.5/5.5 MB 26.8 MB/s eta 0:00:01
----- 5.5/5.5 MB 22.8 MB/s 0:00:00
Using cached pycparser-3.0-py3-none-any.whl (48 kB)
Using cached rfc3339_validator-0.1.4-py2.py3-none-any.whl (3.5 kB)
Downloading tzdata-2026.4-py2.py3-none-any.whl (347 kB)
Using cached uri_template-1.3.0-py3-none-any.whl (11 kB)
Using cached wheel-0.48.0-py3-none-any.whl (33 kB)
Installing collected packages: widgetsnbextension, wheel, websocket-client, webencodings, webcolors, urllib3, uri-template, tzdata, soupsieve, send2trash, rpds-py, rfc3986-validator, rfc3339-validator, pyyaml, pywinpty, python-json-logger, pycryptodome, pycparser, prometheus-client, pandocfilters, mistune, MarkupSafe, lark, jupyterlab_widgets, jupyterlab-pygments, jsonpointer, json5, immutables, idna, h11, fqdn, fastjsonschema, defusedxml, charset_normalizer, certifi, cbor2, babel, attrs, async-lru, vyper, tinycss2, terminado, rfc3987-syntax, requests, referencing, jupyter-builder, jinja2, httpcore, cffi, bleach, beautifulsoup4, arrow, anyio, jupyter-server-terminals, jsonschema-specifications, isoduration, ipywidgets, httpx, cryptography, argon2-cffi-bindings, jupyter-console, jsonschema, argon2-cffi, nbformat, nbclient, jupyter-events, nbconvert, jupyter-server, notebook-shim, jupyterlab-server, jupyter-lsp, jupyterlab, notebook, jupyter

```

```

----- 1/74 [wheel]
- ----- 2/74 [websocket-client]
- ----- 2/74 [websocket-client]
- ----- 2/74 [websocket-client]
- ----- 2/74 [websocket-client]
- ----- 2/74 [websocket-client]
- ----- 2/74 [websocket-client]
-- ----- 4/74 [webcolors]
-- ----- 5/74 [urllib3]
-- ----- 5/74 [urllib3]
-- ----- 5/74 [urllib3]
-- ----- 5/74 [urllib3]
-- ----- 5/74 [urllib3]
-- ----- 5/74 [urllib3]
--- ----- 6/74 [uri-template]
--- ----- 7/74 [tzdata]
--- ----- 8/74 [soupsieve]
----- 9/74 [send2trash]
----- 9/74 [send2trash]
----- 13/74 [pyyaml]

```

[illegible]

```
----- 33/74 [charset_normalizer]
----- 33/74 [charset_normalizer]
----- 34/74 [certifi]
----- 35/74 [cbor2]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 36/74 [babel]
----- 37/74 [attrs]
----- 37/74 [attrs]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 39/74 [vyper]
----- 40/74 [tinycss2]
----- 41/74 [terminado]
----- 43/74 [requests]
----- 43/74 [requests]
----- 43/74 [requests]
----- 44/74 [referencing]
----- 45/74 [jupyter-builder]
----- 45/74 [jupyter-builder]
----- 45/74 [jupyter-builder]
----- 46/74 [jinja2]
----- 46/74 [jinja2]
----- 46/74 [jinja2]
----- 46/74 [jinja2]
----- 46/74 [jinja2]
----- 47/74 [httpcore]
----- 47/74 [httpcore]
----- 47/74 [httpcore]
----- 48/74 [cffi]
----- 48/74 [cffi]
----- 48/74 [cffi]
----- 49/74 [bleach]
----- 49/74 [bleach]
----- 49/74 [bleach]
----- 49/74 [bleach]
----- 49/74 [bleach]
----- 50/74 [beautifulsoup4]
----- 50/74 [beautifulsoup4]
```

```
----- 51/74 [arrow]
----- 51/74 [arrow]
----- 52/74 [anyio]
----- 52/74 [anyio]
----- 52/74 [anyio]
----- 52/74 [anyio]
----- 53/74 [jupyter-server-terminals]
----- 55/74 [isoduration]
----- 56/74 [ipywidgets]
----- 56/74 [ipywidgets]
----- 56/74 [ipywidgets]
----- 56/74 [ipywidgets]
----- 56/74 [ipywidgets]
----- 56/74 [ipywidgets]
----- 57/74 [httpx]
----- 57/74 [httpx]
----- 57/74 [httpx]
----- 58/74 [cryptography]
----- 58/74 [cryptography]
----- 58/74 [cryptography]
----- 58/74 [cryptography]
----- 58/74 [cryptography]
----- 58/74 [cryptography]
----- 60/74 [jupyter-console]
----- 60/74 [jupyter-console]
----- 61/74 [jsonschema]
----- 61/74 [jsonschema]
----- 61/74 [jsonschema]
----- 61/74 [jsonschema]
----- 62/74 [argon2-cffi]
----- 63/74 [nbformat]
----- 63/74 [nbformat]
----- 63/74 [nbformat]
----- 64/74 [nbclient]
----- 65/74 [jupyter-events]
----- 65/74 [jupyter-events]
----- 66/74 [nbconvert]
----- 66/74 [nbconvert]
----- 66/74 [nbconvert]
----- 66/74 [nbconvert]
----- 66/74 [nbconvert]
----- 66/74 [nbconvert]
----- 67/74 [jupyter-server]
----- 67/74 [jupyter-server]
----- 67/74 [jupyter-server]
----- 67/74 [jupyter-server]
----- 67/74 [jupyter-server]
----- 67/74 [jupyter-server]
----- 67/74 [jupyter-server]
----- 67/74 [jupyter-server]
----- 68/74 [notebook-shim]
----- 69/74 [jupyterlab-server]
----- 69/74 [jupyterlab-server]
----- 69/74 [jupyterlab-server]
----- 70/74 [jupyter-lsp]
----- 70/74 [jupyter-lsp]
```

```

----- -- 70/74 [jupyter-lsp]
----- -- 70/74 [jupyter-lsp]
----- - 71/74 [jupyterlab]
----- - 71/74 [jupyterlab]
----- - 71/74 [jupyterlab]
----- - 71/74 [jupyterlab]
----- - 71/74 [jupyterlab]
----- - 71/74 [jupyterlab]
----- - 71/74 [jupyterlab]
----- - 71/74 [jupyterlab]
----- - 72/74 [notebook]
----- - 72/74 [notebook]
----- - 72/74 [notebook]
----- 74/74 [jupyter]

```

Successfully installed MarkupSafe-3.0.3 anyio-4.15.1 argon2-cffi-25.1.0 argon2-cffi-bindings-26.1.0 arrow-1.4.0 async-lru-2.3.0 attrs-26.1.0 babel-2.18.0 beautifulsoup4-4.15.0 bleach-6.4.0 cbor2-5.9.0 certifi-2026.7.22 cffi-2.1.1 charset_normalizer-3.5.1 cryptography-50.0.1 defusedxml-0.7.1 fastjsonschema-2.22.2 fqdn-1.5.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.20 immutables-0.21 ipywidgets-8.1.9 isoduration-20.11.0 jinja2-3.1.6 json5-0.15.0 jsonpointer-3.1.1 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 jupyter-1.1.1 jupyter-builder-1.2.3 jupyter-console-6.6.3 jupyter-events-0.12.1 jupyter-lsp-2.3.1 jupyter-server-2.21.1 jupyter-server-terminals-0.5.4 jupyterlab-4.6.4 jupyterlab-pygments-0.3.0 jupyterlab-server-2.28.1 jupyterlab_widgets-3.0.17 lark-1.3.1 mistune-3.3.4 nbclient-0.11.0 nbconvert-7.17.1 nbformat-5.11.1 notebook-7.6.3 notebook-shim-0.2.4 pandocfilters-1.5.1 prometheus-client-0.26.0 pycparser-3.0 pycryptodome-3.23.0 python-json-logger-4.2.0 pywinpty-3.0.5 pyyaml-6.0.3 referencing-0.37.0 requests-2.34.2 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.6.3 send2trash-2.1.0 soupsieve-2.9.2 terminado-0.18.1 tinycss2-1.5.1 tzdata-2026.4 uri-template-1.3.0 urllib3-2.8.0 vyper-0.4.3 webcolors-25.10.0 webencodings-0.6.1 websocket-client-1.9.2 wheel-0.48.0 widgetsnbextension-4.0.16

Note: you may need to restart the kernel to use updated packages.

In [3]: `import importlib.metadata as md`

```

packages = ['cryptography', 'vyper', 'jupyter', 'ipykernel']
print(f'{"Package":<16} Version')
print('-' * 28)
for name in packages:
    try:
        print(f'{name:<16} {md.version(name)}')
    except md.PackageNotFoundError:
        print(f'{name:<16} CHUA CAI')

```

Package	Version
cryptography	50.0.1
vyper	0.4.3
jupyter	1.1.1
ipykernel	7.3.0

4. (Tuỳ chọn) OpenSSL trên Windows

Không bắt buộc - notebook dùng `cryptography` tự sinh khóa.

```
choco install openssl
openssl genrsa -out ty.pem 2048
openssl rsa -in ty.pem -pubout -out ty.pub
Hoặc tải file .exe : https://github.com/openssl/openssl/wiki/Binaries
```

```
In [ ]: import shutil
import subprocess

openssl = shutil.which('openssl')
if openssl:
    out = subprocess.check_output([openssl, 'version'], text=True).strip()
    print('[OK] OpenSSL:', out)
    print(' Path :', openssl)
else:
    print('[SKIP] OpenSSL chưa có trong PATH (dùng cryptography cung duoc).')
```

5. Đăng ký kernel Jupyter cho .venv

```
In [4]: import sys
import subprocess

cmd = [
    sys.executable,
    '-m',
    'ipykernel',
    'install',
    '--user',
    '--name=is355-venv',
    '--display-name=Python (IS355 .venv)',
]
print('Running:', ' '.join(cmd))
subprocess.check_call(cmd)
print()
print('[OK] Kernel đã đăng ký: Python (IS355 .venv)')
print(' Cursor: chọn kernel này ở góc trên bên phải notebook.')
```

Running: d:\Study\IS355\.venv\Scripts\python.exe -m ipykernel install --user --name=is355-venv --display-name=Python (IS355 .venv)

[OK] Kernel đã đăng ký: Python (IS355 .venv)
Cursor: chọn kernel này ở góc trên bên phải notebook.

6. Smoke test

Kiểm tra: SHA-256 (blockchain), chữ ký số (cryptography), biên dịch Vyper.

```
In [5]: import hashlib
import json

from cryptography.hazmat.backends import default_backend
from cryptography.hazmat.primitives import hashes
```

```

from cryptography.hazmat.primitives.asymmetric import padding, rsa

# --- Blockchain hash ---
payload = json.dumps({'id': 1, 'history': 'Ty likes cat'}, sort_keys=True).encode()
digest = hashlib.sha256(payload).hexdigest()
print('SHA-256 sample:', digest[:32], '...')

# --- Digital signature ---
private_key = rsa.generate_private_key(65537, 2048, default_backend())
message = b'Ty likes cat'
signature = private_key.sign(
    message,
    padding.PSS(mgf=padding.MGF1(hashes.SHA256()), salt_length=padding.PSS.MAX_LENGTH,
    hashes.SHA256()),
)
private_key.public_key().verify(
    signature,
    message,
    padding.PSS(mgf=padding.MGF1(hashes.SHA256()), salt_length=padding.PSS.MAX_LENGTH,
    hashes.SHA256()),
)
print('[OK] cryptography (sign/verify)')

# --- Vyper compile ---
import vyper
from vyper.compiler import compile_code

sample = (
    '# @version ^0.4.0\n'
    '\n'
    'number: uint256\n'
    '\n'
    '@external\n'
    'def store(num: uint256):\n'
    '    self.number = num\n'
    '\n'
    '@external\n'
    'def retrieve() -> uint256:\n'
    '    return self.number\n'
)
result = compile_code(sample)
bytecode = result.get('bytecode') or ''
print('Vyper version:', vyper.__version__)
print('Bytecode (dau):', str(bytecode)[:48], '...')
print('[OK] vyper compile')
print('[OK] Setup xong - keo xuong phan bai tap ly thuyet ben duoi.')

```

SHA-256 sample: 1edbb2e508758f92a2e435fa20226128 ...

[OK] cryptography (sign/verify)

Vyper version : 0.4.3

Bytecode (dau): 0x61004361000f6000396100436000f35f3560e01c636057 ...

[OK] vyper compile

[OK] Setup xong - keo xuong phan bai tap ly thuyet ben duoi.

A. Blockchain

Mỗi khối chứa nhiều giao dịch chuyển giá trị (hoặc thông tin khác) giữa những người tham gia, được bảo mật bằng **mật mã học (cryptography)**. Một sự **đồng thuận (consensus)** giữa nhiều **nút (nodes)** nắm giữ cùng một cơ sở dữ liệu sẽ quyết định khối mới nào được thêm tiếp theo.

- Không thể thay đổi dữ liệu đã ghi.
- Minh bạch và có thể kiểm chứng.
- Được bảo mật nhờ mật mã học và cơ chế đồng thuận.

	Bitcoin	Ethereum
Nội dung khối	Chủ yếu lưu giao dịch (vd: Tỷ gửi 1 BTC cho Tèo)	Giao dịch + thay đổi trạng thái (change of state)
Ví dụ trạng thái	—	Hàng đợi mua vé: trống hoặc đầy

Chuỗi khối (chain)

```

Block A (ID=1, Parent=-) → Block B (ID=2, Parent=1) → Block C
(ID=3, Parent=2)
    Tx1, Tx2, Tx3                Tx4, Tx5                Tx6,
Tx7

```

Mỗi khối lưu `parent_id` và `parent_hash` (SHA-256 của khối cha) để tạo liên kết bất biến.

```
In [6]: import hashlib
import json

class Block:
    id = None
    history = None
    parent_id = None
    parent_hash = None

def block_hash(block: Block) -> str:
    """Tính SHA-256 của toàn bộ thuộc tính khối."""
    payload = json.dumps(block.__dict__, sort_keys=True).encode("utf-8")
    return hashlib.sha256(payload).hexdigest()

block_A = Block()
block_A.id = 1
block_A.history = "Ty likes cat"

block_B = Block()
block_B.id = 2
block_B.history = "Teo likes dog"
block_B.parent_id = block_A.id
block_B.parent_hash = block_hash(block_A)

block_C = Block()
block_C.id = 3
block_C.history = "Mai likes dog"
block_C.parent_id = block_B.id
block_C.parent_hash = block_hash(block_B)

chain = [block_A, block_B, block_C]
for b in chain:
    print(f"Block {b.id}")
    print(f"  history      : {b.history}")
    print(f"  parent_id     : {b.parent_id}")
    print(f"  parent_hash   : {b.parent_hash}")
    print(f"  self_hash     : {block_hash(b)}")
    print()
```

```

Block 1
  history      : Ty likes cat
  parent_id    : None
  parent_hash  : None
  self_hash    : 1edbb2e508758f92a2e435fa20226128dac7499ca3f14b9c1d949329e24dc5b7

Block 2
  history      : Teo likes dog
  parent_id    : 1
  parent_hash  : 1edbb2e508758f92a2e435fa20226128dac7499ca3f14b9c1d949329e24dc5b7
  self_hash    : 46bed02f0ca83b09c4df73e2e7efe66b8331a0bfdceba8e94709b3c6329b4db

Block 3
  history      : Mai likes dog
  parent_id    : 2
  parent_hash  : 46bed02f0ca83b09c4df73e2e7efe66b8331a0bfdceba8e94709b3c6329b4db
  self_hash    : 811aebc35da972c4693f5595b49ce6c302933aa4f798a4512f88dbf1733d8323

```

Thử nghiệm: sửa dữ liệu khối cha → hash không còn khớp

Nếu thay đổi `history` của Block A, `parent_hash` trong Block B sẽ **không còn đúng** — minh họa tính bất biến của blockchain.

```

In [7]: # Giả sử kẻ tấn công sửa Block A
        block_A.history = "Ty hates cat"

        expected = block_B.parent_hash
        actual = block_hash(block_A)

        print("parent_hash lưu trong Block B:", expected)
        print("hash Block A sau khi bị sửa :", actual)
        print("Chuỗi còn hợp lệ?", expected == actual)

```

```

parent_hash lưu trong Block B: 1edbb2e508758f92a2e435fa20226128dac7499ca3f14b9c1d949
329e24dc5b7
hash Block A sau khi bị sửa : 19a6fedcc944890d7ecf80470f3e93b5bc2f8e0c6ba5d17883f10
6b56e3efd9d
Chuỗi còn hợp lệ? False

```

B. Chữ ký số lên dữ liệu trong blockchain

Trong blockchain dùng **hai khóa** để ký số dữ liệu: xác thực thông điệp và bảo vệ khỏi bị sửa bởi người trái phép.

Khóa	Vai trò
Private key (khóa riêng)	Giữ bí mật; dùng để ký thông điệp
Public key (khóa công khai)	Công khai; dùng để xác minh chữ ký

Tạo khóa bằng OpenSSL (tuỳ chọn, Windows)

Cài OpenSSL bằng Chocolatey, hoặc tải `.exe` tại <https://github.com/openssl/openssl/wiki/Binaries>

```
choco install openssl
openssl genrsa -out ty.pem 2048
openssl rsa -in ty.pem -pubout -out ty.pub
```

Trong notebook bên dưới, ta dùng thư viện `cryptography` của Python để **tự sinh khóa** (không bắt buộc OpenSSL).

Cài thư viện

```
In [8]: # Chạy 1 lần nếu chưa có cryptography
%pip install cryptography -q
```

Note: you may need to restart the kernel to use updated packages.

TySignature — cấu hình, ký và xác minh

Tương đương file `TySignature.py` trong bài giảng.

```
In [9]: from pathlib import Path

from cryptography.hazmat.backends import default_backend
from cryptography.hazmat.primitives import hashes, serialization
from cryptography.hazmat.primitives.asymmetric import padding, rsa

# === Configuration ===
GENERATE_PRIVATE_KEY = True
DERIVE_PUBLIC_KEY_FROM_PRIVATE_KEY = True
PRIVATE_KEY_FILE = "ty.pem"
PUBLIC_KEY_FILE = "ty.pub"
MESSAGE = b"Ty likes cat"

# --- Private key ---
if GENERATE_PRIVATE_KEY:
    private_key = rsa.generate_private_key(
        public_exponent=65537,
        key_size=2048,
        backend=default_backend(),
    )
    Path(PRIVATE_KEY_FILE).write_bytes(
        private_key.private_bytes(
            encoding=serialization.Encoding.PEM,
            format=serialization.PrivateFormat.TraditionalOpenSSL,
            encryption_algorithm=serialization.NoEncryption(),
        )
    )
    print(f"Đã tạo private key → {PRIVATE_KEY_FILE}")
else:
```

```

private_key = serialization.load_pem_private_key(
    Path(PRIVATE_KEY_FILE).read_bytes(),
    password=None,
    backend=default_backend(),
)
print(f"Đã nạp private key từ {PRIVATE_KEY_FILE}")

# --- Sign ---
signature = private_key.sign(
    MESSAGE,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH,
    ),
    hashes.SHA256(),
)

print("MESSAGE :", MESSAGE)
print("SIGNATURE (hex):", signature.hex()[:64], "...")
print("SIGNATURE length:", len(signature), "bytes")

```

Đã tạo private key → ty.pem

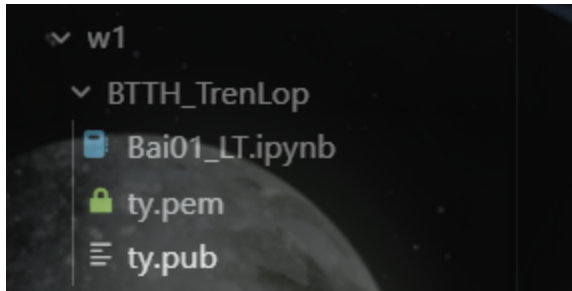
MESSAGE : b'Ty likes cat'

SIGNATURE (hex): 1683f62dc5884a81d809c023e923b606e2794697d04e69594da15e99c12cc1f4

...

SIGNATURE length: 256 bytes

Kết quả: file khóa ty.pem / ty.pub đã tạo trong thư mục bài.



```
Bai01_LT.ipynb  ty.pem X
TH > w1 > BTTH_TrenLop > ty.pem
1 -----BEGIN RSA PRIVATE KEY-----
2 MIIEowIBAAKCAQEA8nYFDs2c3jsRS8HYJa2ogAnMDWQe2w96Al/xybcnsgtzgx4X
3 0lQggqgOAFI+GgN87qToBUNuoABOJA8NDTqKr9r8vhelc2M+LbiXYLXrcrNBZh4X
4 uQ5J5P7U4Gx0xEiUMBA4lpP0iGkcCAYlhdWFE8lmgqrZ9xcTqXTJuVPfe/giohz0
5 Kne0SpU3eyO2uhfn+1FferNfEAaLkRDNDs5p0zuC7UiKh5+qwvYAXLm1UzUfgQnY
6 s74uTUnqpr8KYrdP6IeIT1sSZpHG66txL3b3+EJxQ175dzqWoTv/zjjPeNgv5Zc6
7 wuOLtyKKfao/2zx31TR3v1q3P4HNSic12HS/ewIDAQABaoIBACVfb2M16r1Poy4a
8 uIxXlW1QVUghIn0bXcVSRazku9SQuFh45keMjLW2dCPUZGP774XOX5fiQSXrMke
9 cCFrCyfqe40k4Zyvc6JM3Y82K2SC4KCdddmrAQ6CtUhp+1Tff1fRWG3Ik5efp12
10 rmpP4b3i4ZlgPXh1Rw4WkZDZU2RwpzWculOrGyYr+fc/y+jdVg5eJ/6IToh179j
11 3+R0845Nm6vXh1kRwmQ7RwcGUp04S36D20c37t/O+2ovEk76b4RgaQ1I4iq5GtTK
12 AACne3xcqF1I8sdINK3398f1uSiDyWfMKwf2Vmu9ScuSQXx1nNj+DR0iSPXbjwL
13 cFClLUEGgYEA+wIJCaUuz4cIR9qKB4mgT7TyDdxNTtcJgMSran2SLj5726TuDvQ0
14 2QBRthlobUHglGO3PaXxlb/ymqR3SuucuejrEFBIuL1SM4ozcK4KrYRgfX05Mayu
15 +I759Kit0Zs00yJD2fgc4cYsM9BY1lChrvGrQq8Ve/qIVpYQxwMeRrsCgYEA90h4
16 G4nCeV/zHm/VDhU/C3MIodGe9qicBPfVTxme//SL30BmQwxVZeJYLoPFMBocrhsD
17 sXEcJWeoMsXU5xYK0sgZ/mex1D2ThP08u4MwWoX9cmZW9hBfADQZ4PFuqqE7rGc
18 S9jzOEX+WLe/sI71PtMLa5Az/gviwHWPk+JyvKEGyBwJJilFoezcEDi2WCRvYtC
19 ETWTu7lOYAR+418yVKfawU/U8frp83EpAlt/YA8KAUS1B0Npp5onrXpW0RvNJTW
20 +n8K6tOSTgljw/r3fMqIe3wgkHNNHjmOE1LZWRvfaccR5F0sJQ3CUSEufocU51VW
21 M+OwB0kx+tyu0pzsTrhZwQKBgQDHin8BGCUIVFjGss9lsddR0d2hVpnDyIM9rw89
22 6RrYx+kmmbzrVcSDloY54rL4AcTb9WihgviHZoHflqwaXTsWo8n19I7jZFzp779W
23 QKKAKac+7T43CTbf3yPBkD+6RwdHhAltWx9iZXi89mmEe5eCvqJ0hcSut8CeNT9K
24 wmx9AQKBgDn4jYimwXWRrF9BOM9BK7Jzafileymve3upHH+sMzhbZ34tipxoiNytN
25 00BU1k0rUwTv1pzvmiU8dfixximUYlCVhJQdvQekIvLFvU1VBcx5iy/V9aMNRxK
26 +SctXSAmhBXvPm1k+SS/LqIFihNv1E5o5fP92lVvmA+emfBwtyS+
27 -----END RSA PRIVATE KEY-----
28
```

```
Bai01_LT.ipynb  ty.pub X
TH > w1 > BTTH_TrenLop > ty.pub
1 -----BEGIN PUBLIC KEY-----
2 MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA8nYFDs2c3jsRS8HYJa2o
3 gAnMDWQe2w96Al/xybcnsgtzgx4X0lQggqgOAFI+GgN87qToBUNuoABOJA8NDTqK
4 r9r8vhelc2M+LbiXYLXrcrNBZh4XuQ5J5P7U4Gx0xEiUMBA4lpP0iGkcCAYlhdWF
5 E8lmgqrZ9xcTqXTJuVPfe/giohz0Kne0SpU3eyO2uhfn+1FferNfEAaLkRDNDs5p
6 OzuC7UiKh5+qwvYAXLm1UzUfgQnYs74uTUnqpr8KYrdP6IeIT1sSZpHG66txL3b3
7 +EJxQ175dzqWoTv/zjjPeNgv5Zc6wuOLtyKKfao/2zx31TR3v1q3P4HNSic12HS/
8 ewIDAQAB
9 -----END PUBLIC KEY-----
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
```

```
In [10]: from pathlib import Path

# --- Public key ---
if DERIVE_PUBLIC_KEY_FROM_PRIVATE_KEY:
    public_key = private_key.public_key()
```

```

Path(PUBLIC_KEY_FILE).write_bytes(
    public_key.public_bytes(
        encoding=serialization.Encoding.PEM,
        format=serialization.PublicFormat.SubjectPublicKeyInfo,
    )
)
print(f"Đã derive + lưu public key → {PUBLIC_KEY_FILE}")
else:
    public_key = serialization.load_pem_public_key(
        Path(PUBLIC_KEY_FILE).read_bytes(),
        backend=default_backend(),
    )
    print(f"Đã nạp public key từ {PUBLIC_KEY_FILE}")

# --- Verify ---
try:
    public_key.verify(
        signature,
        MESSAGE,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH,
        ),
        hashes.SHA256(),
    )
    print("✓ Xác minh thành công: chữ ký hợp lệ với MESSAGE gốc.")
except Exception as e:
    print("✗ Xác minh thất bại:", e)

```

Đã derive + lưu public key → ty.pub

✓ Xác minh thành công: chữ ký hợp lệ với MESSAGE gốc.

Yêu cầu thực hành tại lớp (mục B)

1. **Vai Tèo:** cố sửa thông điệp của Tý thành "Ty hates cat" rồi kiểm tra chữ ký.
2. Chạy lại với thông điệp gốc "Ty likes cat", chụp chữ ký; đổi thành "Ty hates cat", chạy lại và **so sánh** hai chữ ký.
3. (*) Viết chương trình xác thực xem có phải Tý đã viết "Ty likes cat" hay không.

In [11]: *# Yêu cầu 1 & 2: Tèo sửa thông điệp → chữ ký không còn hợp lệ*
TAMPERED = b"Ty hates cat"

```
print("Chữ ký gốc (64 hex đầu):", signature.hex()[:64])
```

Ký Lại với thông điệp đã sửa (nếu Tý ký Lại) → chữ ký khác hoàn toàn

```

sig_tampered = private_key.sign(
    TAMPERED,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH,
    ),
    hashes.SHA256(),
)
print("Chữ ký mới (Ty hates cat):", sig_tampered.hex()[:64])

```

```

print("Hai chữ ký giống nhau?", signature == sig_tampered)

# Tòe giữ chữ ký gốc nhưng đổi nội dung → verify FAIL
try:
    public_key.verify(
        signature,
        TAMPERED,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH,
        ),
        hashes.SHA256(),
    )
    print("Verify với thông điệp đã sửa: PASS (không mong đợi!)")
except Exception:
    print("Verify với thông điệp đã sửa: FAIL ← đúng như kỳ vọng (toàn vẹn bị phá)")

```

Chữ ký gốc (64 hex đầu): 1683f62dc5884a81d809c023e923b606e2794697d04e69594da15e99c12cc1f4

Chữ ký mới (Ty hates cat): 1a497afd600b5e541d103510f2dc33a7b2add80fc4d2544d1b12932e46c35fdb

Hai chữ ký giống nhau? False

Verify với thông điệp đã sửa: FAIL ← đúng như kỳ vọng (toàn vẹn bị phá)

In [12]: # Yêu cầu 3 (*): xác thực Tý có phải người viết "Ty likes cat" không?

```

def verify_ty_message(message: bytes, sig: bytes, pub_key) -> bool:
    try:
        pub_key.verify(
            sig,
            message,
            padding.PSS(
                mgf=padding.MGF1(hashes.SHA256()),
                salt_length=padding.PSS.MAX_LENGTH,
            ),
            hashes.SHA256(),
        )
        return True
    except Exception:
        return False

claimed = b"Ty likes cat"
ok = verify_ty_message(claimed, signature, public_key)
print(f"Thông điệp: {claimed!r}")
print("Có phải Tý (chủ private key) đã ký?", "CÓ ✓" if ok else "KHÔNG X")

```

Thông điệp: b'Ty likes cat'

Có phải Tý (chủ private key) đã ký? CÓ ✓

C. Smart contract

Smart contract (hợp đồng thông minh) là chương trình sống trên blockchain, giải quyết hạn chế của Bitcoin (chỉ giao dịch tài chính).

Bitcoin		Ethereum / Smart contract
Giao dịch	Chuyển giá trị (vd: Tý gửi 1 BTC cho Tèo)	Thay đổi trạng thái chương trình (vd: biến từ 5 → 9)
Bản chất	Tài chính	State transition của chương trình

Cách hiểu trực quan

- Smart contract $\approx$ file nhị phân (như `.exe` trên Windows) nhưng **sống trên blockchain**.
- Chương trình tồn tại dưới dạng **bytecode dựa trên stack**.
- Khi tương tác, người dùng gửi lệnh → **EVM (Ethereum Virtual Machine)** đẩy/kéo giá trị từ stack và thực thi.
- Khác ứng dụng web: bytecode smart contract **minh bạch**, mọi nút đều giữ bản sao giống nhau.

Hình 2. Smart contract tồn tại trên mọi node trong chuỗi khối.

Hình 3. Người dùng gửi input (instruction) tới smart contract.

D. Sử dụng Vyper hiện thực một hợp đồng thông minh

i. Môi trường triển khai trên Ethereum (Remix)

1. Mở trình duyệt → <https://remix.ethereum.org>
2. Trong **FILE EXPLORER**, mở thư mục `contracts`.
3. Click phải file `1_Storage.sol` → **Compile**.
4. Vào **Deploy & run transactions** → Deploy lên môi trường Remix VM / testnet.
5. Kiểm tra kết quả triển khai (địa chỉ hợp đồng, các hàm `store` / `retrieve`).

ii. Thiết lập Vyper

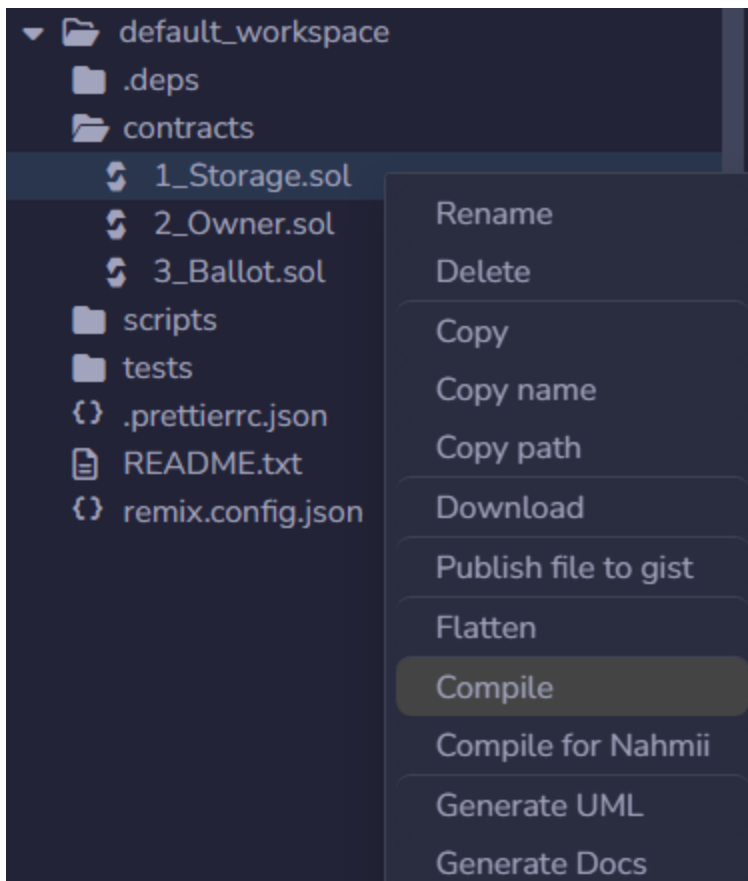
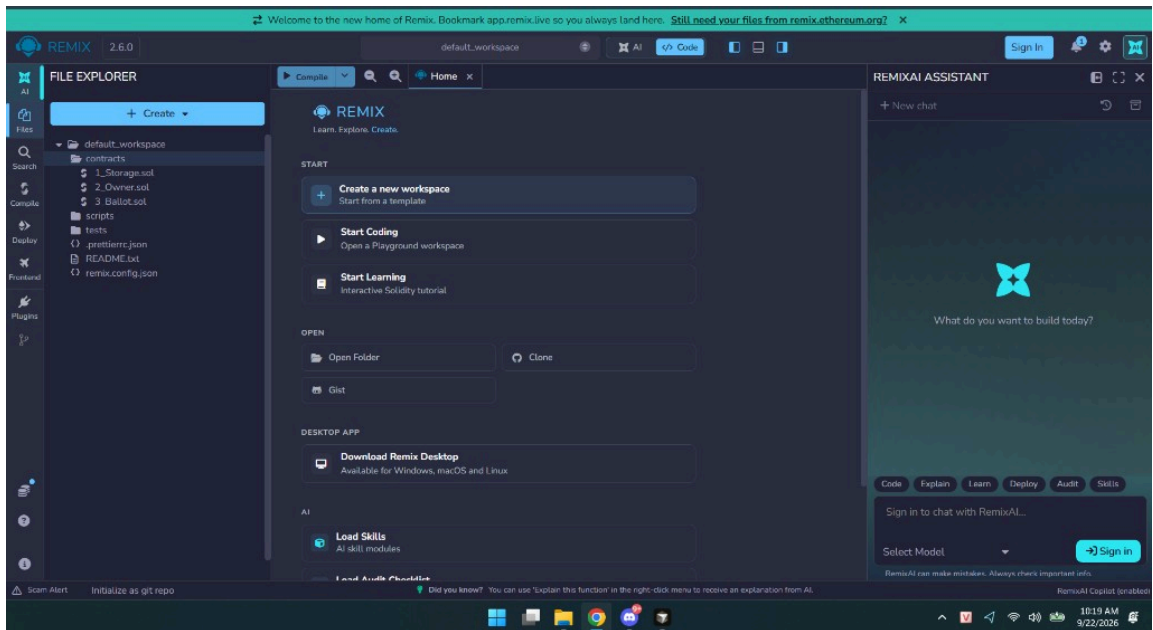
Trên Remix (web):

1. Mở **Plugin Manager** → tìm `vyper` → Activate.
2. Xuất hiện icon compiler Vyper trên sidebar.
3. Chọn file `.vy` → Compile.

iii. Hợp đồng đơn giản đầu tiên — `Storage.vy`

Biên dịch / Deploy trên **Remix** (plugin Vyper) — không cần cài Vyper local.

Kết quả Remix: biên dịch và triển khai `1_Storage.sol` trên Remix VM.



AI

Files

Search

Compile

Deploy

Frontend

Plugins

DEPLOY & RUN TRANSACTIONS

Environment

Fork

Reset

Remix VM

Osaka

Account 1

0x5B3...eddC4

100.000 ETH

Deploy

Deployed contracts

Transactions history

Deploy

Remix VM osaka

Storage

1_Storage.sol

Compiled

Value

0

wei

Gas limit

auto

0

Deploy

Deploy

Deployed contracts 1

Transactions history 1

Deployed Contracts 1

+ Add Contract

Interact with a deployed contract

Storage

0xd91...39138

Remix VM

2s ago

Functions

● store (uint256)

num (uint256)

Transact

Calldata

Params

● retrieve

Call

Low level interaction

Deploy

Deployed contracts 1

Transactions history 1

Transactions recorder 1

Save

Contract call

Transaction list

Newest

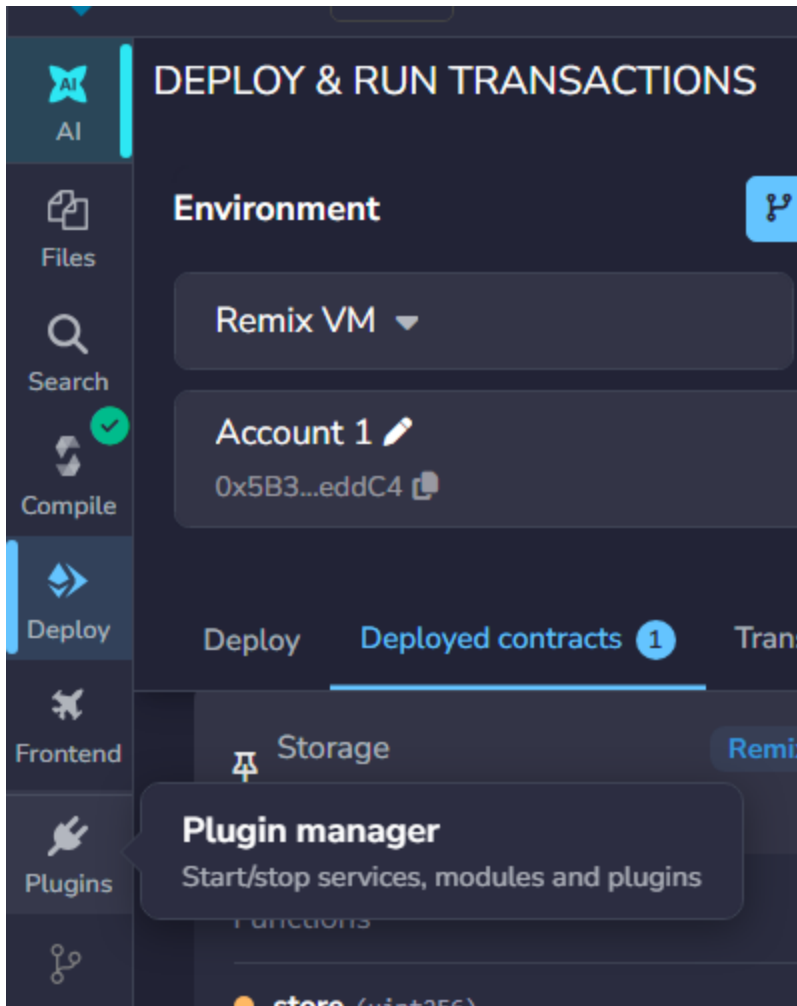
Storage

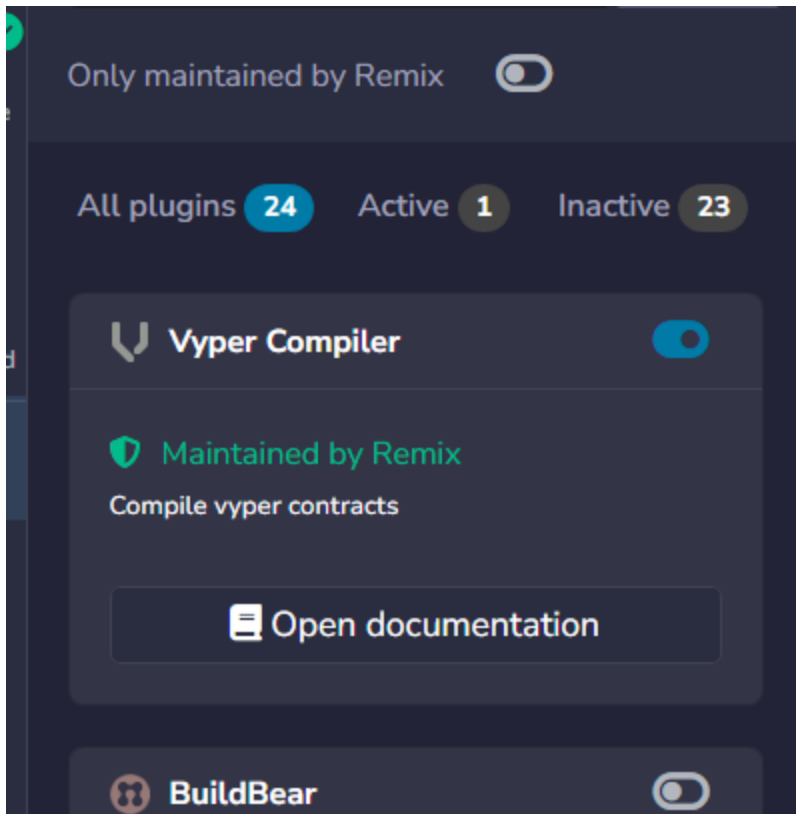
0xd91...39138

1 transaction

Deployed 0s ago

Kết quả thiết lập Vyper trên Remix:





Mã nguồn `Storage.vy` (Hình 10)

Sau mỗi cell ghi file `.vy`, chạy **cell ngay bên dưới** để biên dịch và in kết quả (ABI + bytecode).

```
In [13]: from pathlib import Path

# Nội dung file Storage.vy – chạy cell này để ghi file ra thư mục hiện tại
storage_vy = '''# @version ^0.4.3

number: uint256

@external
def store(num: uint256):
    self.number = num

@external
def retrieve() -> uint256:
    return self.number
'''

Path("Storage.vy").write_text(storage_vy, encoding="utf-8")
print("Đã ghi Storage.vy")
```

```

Đã ghi Storage.vy
# @version ^0.4.3

number: uint256

@external
def store(num: uint256):
    self.number = num

@external
def retrieve() -> uint256:
    return self.number

```

```

In [21]: # Ket qua bien dich: Storage.vy
from pathlib import Path
from vyper.compiler import compile_code

_src = Path("Storage.vy").read_text(encoding="utf-8")
print("File:", Path("Storage.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "function"]
    bytecode = str(result.get("bytecode") or "")
    print("[OK] Bien dich thanh cong")
    print("ABI functions:", ", ".join(fns) if fns else "(khong co)")
    print("Bytecode (64 ky tu dau):", bytecode[:64], ("..." if len(bytecode) > 64 else ""))
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)

```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\Storage.vy

```

-----
[OK] Bien dich thanh cong
ABI functions: store, retrieve
Bytecode (64 ky tu dau): 0x61004361000f6000396100436000f35f3560e01c636057361d8118610
02057 ...
Bytecode length: 270 chars

```

E. Ngôn ngữ lập trình Vyper

i. Kiểu dữ liệu

Nhóm	Ví dụ
Số nguyên có dấu	int8, int64, int128, int256
Số nguyên không dấu	uint8, uint64, uint128, uint256
Boolean	bool

Nhóm	Ví dụ
Địa chỉ	<code>address</code>
Mảng kích thước cố định	<code>int16[5]</code>
Mảng Byte	<code>bytes32</code> , <code>Bytes[56]</code>
Chuỗi	<code>String[100]</code>
Enum / Flag	<code>flag Direction: ...</code>
List / DynArray	<code>DynArray[int128, 5]</code>
Struct	<code>struct Permission: ...</code>
Mapping	<code>HashMap[address, uint256]</code>

`DataType.vy` (Hình 13) — khai báo các kiểu

```
In [14]: from pathlib import Path

datatype_vy = '''# @version ^0.4.3

life_is_beautiful: bool

var_int1: int8
var_int2: int64
var_int3: int128
var_int4: int256

var_uint1: uint8
var_uint2: uint64
var_uint3: uint128
var_uint4: uint256

my_grandma_wallet: address

var_byte1: bytes32
var_byte2: bytes18
var_bytes: Bytes[56]

author: String[100]

flag Direction:
    NORTH
    SOUTH
    WEST
    EAST

direction: Direction

my_list: int16[5]
my_dynamic_array: DynArray[int128, 5]
```

```

struct Permission:
    write: bool
    execute: bool

my_permission: Permission
donaturs: HashMap[address, uint256]
...

Path("DataType.vy").write_text(datatype_vy, encoding="utf-8")
print("Da ghi DataType.vy")

```

Đã ghi DataType.vy

```

In [22]: # Ket qua bien dich: DataType.vy
from pathlib import Path
from vyper.compiler import compile_code

_src = Path("DataType.vy").read_text(encoding="utf-8")
print("File:", Path("DataType.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "
    bytecode = str(result.get("bytecode") or "")
    print("[OK] Bien dich thanh cong")
    print("ABI functions:", ", ".join(fns) if fns else "(khong co)")
    print("Bytecode (64 ky tu dau):", bytecode[:64], ("..." if len(bytecode) > 64 e
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)

```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\DataType.vy

```

-----
[OK] Bien dich thanh cong
ABI functions: (khong co)
Bytecode (64 ky tu dau): 0x61000361000f6000396100036000f35f5fffd8558208b8a5669fa721aa
836a7 ...
Bytecode length: 140 chars

```

ii. Hàm (Functions)

Hàm trong Vyper giống Python: nhóm câu lệnh trong một khối và thực thi khi được gọi.

- Ví dụ đã thấy: `store` / `retrieve` trong `Storage.vy` .
- Giống Python, nếu có `__init__` thì hàm này chạy **khi deploy** (constructor).
- Trong Vyper 0.4.x, constructor dùng decorator `@deploy` .

StorageInit.vy (Hình 14) — hàm khởi tạo

```

In [15]: from pathlib import Path

storage_init_vy = '''# @version ^0.4.3

```

```

my_grandma_wallet: address
author: String[100]
fib_list: int16[5]
fib_dynamic_array: DynArray[int128, 5]

struct Permission:
    write: bool
    execute: bool

my_permission: Permission
donaturs: HashMap[address, uint256]

@deploy
def __init__():
    self.my_grandma_wallet = 0xde93510CFa39Ab92BF927399F799DbE71997Ee0b
    self.author = "Arjuna Sky Kok"
    self.fib_list = [1, 1, 2, 3, 5]
    self.fib_dynamic_array.append(1)
    self.fib_dynamic_array.append(1)
    self.fib_dynamic_array.append(2)
    self.my_permission = Permission(write=True, execute=False)
    self.donaturs[self.my_grandma_wallet] = 4000000000000000000
    donation_target: int128 = 8000000000000000000
    ...

Path("StorageInit.vy").write_text(storage_init_vy, encoding="utf-8")
print("Da ghi StorageInit.vy")

```

Đã ghi StorageInit.vy

```

In [23]: # Ket qua bien dich: StorageInit.vy
from pathlib import Path
from vyper.compiler import compile_code

_src = Path("StorageInit.vy").read_text(encoding="utf-8")
print("File:", Path("StorageInit.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "
    bytecode = str(result.get("bytecode") or "")
    print("[OK] Bien dich thanh cong")
    print("ABI functions:", ", ".join(fns) if fns else "(khong co)")
    print("Bytecode (64 ky tu dau):", bytecode[:64], ("..." if len(bytecode) > 64 e
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)

```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\StorageInit.vy

[OK] Biên dịch thành công

ABI functions: (không có)

Bytecode (64 ký tự đầu): 0x346100f75773de93510cfa39ab92bf927399f799dbe71997ee0b5f55600e60 ...

Bytecode length: 612 chars

iii. Mutability (tính biến đổi)

Decorator phạm vi: `@external`, `@internal`.

Decorator **mutability**:

Decorator	Đọc state / env?	Ghi state?	Nhận ETH?
<code>@pure</code>	Không	Không	Không
<code>@view</code>	Có	Không	Không
<code>@nonpayable</code>	Có	Có	Không
<code>@payable</code>	Có	Có	Có

Smart contract có thể hoạt động như **escrow / ngân hàng** nhờ khả năng nhận và giữ ETH.

`Annotations.vy` (Hình 15)

```
In [16]: from pathlib import Path

annotations_vy = '''# @version ^0.4.3

author: String[100]
donatur: String[100]

@deploy
def __init__():
    # đã ngam định @external
    self.author = "Arjuna Sky Kok"

@external
@pure
def add(x: int128, y: int128) -> int128:
    return x + y

@external
@view
def get_name_and_title() -> String[200]:
    return concat("Mr. ", self.author)

@external
@nonpayable
def change_name(new_name: String[100]):
    self.author = new_name
```

```

@external
@payable
def donate(donatur_name: String[100]):
    self.donatur = donatur_name
    ...

Path("Annotations.vy").write_text(annotations_vy, encoding="utf-8")
print("Da ghi Annotations.vy")

```

Đã ghi Annotations.vy

→ Biên dịch & Deploy trên Remix, ghi kết quả vào báo cáo.

```

In [24]: # Ket qua bien dich: Annotations.vy
from pathlib import Path
from vyper.compiler import compile_code

_src = Path("Annotations.vy").read_text(encoding="utf-8")
print("File:", Path("Annotations.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "
bytecode = str(result.get("bytecode") or "")
    print("[OK] Bien dich thanh cong")
    print("ABI functions:", ", ".join(fns) if fns else "(khong co)")
    print("Bytecode (64 ky tu dau):", bytecode[:64], ("..." if len(bytecode) > 64 e
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)

```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\Annotations.vy

[OK] Bien dich thanh cong

ABI functions: add, get_name_and_title, change_name, donate

Bytecode (64 ky tu dau): 0x3461004e57600e6040527f41726a756e6120536b79204b6f6b00000000
00000 ...

Bytecode length: 1386 chars

iv. Cấu trúc điều khiển

Vyper hỗ trợ `if` / `elif` / `else` và vòng `for` giống Python.

```

In [17]: from pathlib import Path

# ControlFlow.vy (Hình 16)
control_vy = '''# @version ^0.4.3

@external
@pure
def sum() -> uint256:
    s: uint256 = 0
    for i: uint256 in [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]:
        s += i

```

```

        return s

@external
@pure
def greet(time: String[10]) -> String[20]:
    if time == "morning":
        return "Good morning!"
    elif time == "evening":
        return "Good evening!"
    else:
        return "How are you?"
...

Path("ControlFlow.vy").write_text(control_vy, encoding="utf-8")
print("Da ghi ControlFlow.vy")

```

Đã ghi ControlFlow.vy

→ Biên dịch & Deploy trên Remix, ghi kết quả vào báo cáo.

```

In [25]: # Ket qua bien dich: ControlFlow.vy
from pathlib import Path
from vyper.compiler import compile_code

_src = Path("ControlFlow.vy").read_text(encoding="utf-8")
print("File:", Path("ControlFlow.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "function"]
    bytecode = str(result.get("bytecode") or "")
    print("[OK] Bien dich thanh cong")
    print("ABI functions:", ", ".join(fns) if fns else "(khong co)")
    print("Bytecode (64 ky tu dau):", bytecode[:64], ("..." if len(bytecode) > 64 else ""))
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)

```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\ControlFlow.vy

```

-----
[OK] Bien dich thanh cong
ABI functions: sum, greet
Bytecode (64 ky tu dau): 0x61027e6100116100003961027e610000f35f3560e01c63853255cc811
86100 ...
Bytecode length: 1418 chars

```

v. Biến môi trường

Biến	Ý nghĩa
block.number	Số thứ tự khối hiện tại
block.timestamp	Timestamp của khối
msg.sender	Địa chỉ người gọi

Biến**Ý nghĩa**

msg.value

Số ETH (wei) gửi kèm giao dịch

```
In [18]: from pathlib import Path

# EnvVar.vy (Hình 17)
envvar_vy = '''# @version ^0.4.3

donatur: address
donation: uint256
time: uint256

@external
@payable
def donate():
    self.donatur = msg.sender
    self.donation = msg.value
    self.time = block.timestamp
...

Path("EnvVar.vy").write_text(envvar_vy, encoding="utf-8")
print("Đã ghi EnvVar.vy")
```

Đã ghi EnvVar.vy

→ Biên dịch & Deploy trên Remix, ghi kết quả vào báo cáo.

```
In [26]: # Ket qua bien dich: EnvVar.vy
from pathlib import Path
from vyper.compiler import compile_code

_src = Path("EnvVar.vy").read_text(encoding="utf-8")
print("File:", Path("EnvVar.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "function"]
    bytecode = str(result.get("bytecode") or "")
    print("[OK] Bien dich thanh cong")
    print("ABI functions:", ", ".join(fns) if fns else "(khong co)")
    print("Bytecode (64 ky tu dau):", bytecode[:64], ("..." if len(bytecode) > 64 else ""))
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)
```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\EnvVar.vy

[OK] Bien dich thanh cong

ABI functions: donate

Bytecode (64 ky tu dau): 0x61002061000f6000396100206000f35f3560e01c63ed88c68e811861001c57 ...

Bytecode length: 200 chars

vi. Event logging

Ngoài biến trạng thái, có thể **ghi nhật ký sự kiện** (log) để theo dõi / thông báo off-chain (vd: đạt mục tiêu quyên góp → đăng mạng xã hội).

```
In [19]: from pathlib import Path

# EventLogging.vy (Hình 18)
event_vy = '''# @version ^0.4.3

event Donation:
    donatur: indexed(address)
    amount: uint256

@external
@payable
def donate():
    log Donation(msg.sender, msg.value)
...

Path("EventLogging.vy").write_text(event_vy, encoding="utf-8")
print("Đã ghi EventLogging.vy")
```

Đã ghi EventLogging.vy

→ Biên dịch & Deploy trên Remix, ghi kết quả vào báo cáo.

```
In [27]: # Ket qua bien dich: EventLogging.vy
from pathlib import Path
from vyper.compiler import compile_code

_src = Path("EventLogging.vy").read_text(encoding="utf-8")
print("File:", Path("EventLogging.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "function"]
    bytecode = str(result.get("bytecode") or "")
    print("[OK] Biên dịch thành công")
    print("ABI functions:", ", ".join(fns) if fns else "(không có)")
    print("Bytecode (64 ký tự đầu):", bytecode[:64], ("..." if len(bytecode) > 64 else ""))
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)
```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\EventLogging.vy

```
-----  
[OK] Bien dich thanh cong  
ABI functions: donate  
Bytecode (64 ky tu dau): 0x61004061000f6000396100406000f35f3560e01c63ed88c68e8118610  
03c57 ...  
Bytecode length: 264 chars  
D:\Study\IS355\.venv\Lib\site-packages\vyper\semantics\types\user.py:313: Deprecatio  
n: Instantiating events with positional arguments is deprecated as of v0.4.1 and wil  
l be disallowed in a future release. Use kwargs instead e.g.:  
...  
log Donation(donatur=msg.sender, amount=msg.value)  
...
```

```
function "donate", line 10:8  
      9 def donate():  
----> 10     log Donation(msg.sender, msg.value)  
-----^  
      11  
  
vyper_warn(Deprecation(msg, node))
```

vii. Interface

Smart contract có thể gọi smart contract khác qua **interface**.

Ví dụ: `StorageClient.vy` gọi hàm `retrieve()` của `Storage.vy` đã deploy trước đó.

```
In [20]: from pathlib import Path  
  
# StorageClient.vy (Hình 19)  
client_vy = '''# @version ^0.4.3  
  
interface Storage:  
    def retrieve() -> uint256: view  
  
storage_contract: Storage  
  
@deploy  
def __init__(storage_address: address):  
    self.storage_contract = Storage(storage_address)  
  
@external  
def call_retrieve() -> uint256:  
    return staticcall self.storage_contract.retrieve()  
...  
  
Path("StorageClient.vy").write_text(client_vy, encoding="utf-8")  
print("Đã ghi StorageClient.vy")
```

Đã ghi `StorageClient.vy`

→ Deploy `Storage.vy` trước, lấy địa chỉ, rồi Deploy `StorageClient` với địa chỉ đó.

```
In [28]: # Ket qua bien dich: StorageClient.vy  
from pathlib import Path
```

```

from vyper.compiler import compile_code

_src = Path("StorageClient.vy").read_text(encoding="utf-8")
print("File:", Path("StorageClient.vy").resolve())
print("-" * 50)
try:
    result = compile_code(_src, output_formats=["abi", "bytecode"])
    abi = result.get("abi") or []
    fns = [x.get("name") for x in abi if isinstance(x, dict) and x.get("type") == "function"]
    bytecode = str(result.get("bytecode") or "")
    print("[OK] Bien dich thanh cong")
    print("ABI functions:", ", ".join(fns) if fns else "(khong co)")
    print("Bytecode (64 ky tu dau):", bytecode[:64], ("..." if len(bytecode) > 64 else ""))
    print("Bytecode length:", len(bytecode), "chars")
except Exception as e:
    print("[FAIL]", type(e).__name__)
    print(e)

```

File: D:\Study\IS355\TH\w1\BTTH_TrenLop\StorageClient.vy

```

-----
[OK] Bien dich thanh cong
ABI functions: call_retrieve
Bytecode (64 ky tu dau): 0x3461002d5760206100b25f395f518060a01c61002d576040526040515
f5561 ...
Bytecode length: 358 chars

```

Kết quả: thư mục BTTH_TrenLop sau khi chạy notebook (các file .vy , ty.pem , ty.pub).

